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"2-nil or not 2-nil, that is the question": a characterization of two-step nilpotent varieties

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What if William Shakespeare had been a mathematician? His most famous quote might have been: "2-nil or not 2-nil?" The aim of my talk is to answer this mathematical variant of Hamlet's question. To do so, I will present a theorem that gives a precise characterization of two-step nilpotent varieties in the sense of universal algebra.

In the abelian case, which amounts to one-step nilpotency, the corresponding answer is provided by Freyd's theorem: abelian varieties are, up to equivalence, precisely categories of modules over rings [3]. Our theorem gives a quadratic counterpart: a finitary, single-sorted variety is two-step nilpotent if and only if it is equivalent to the category of right modules over a square ring [4].

Here, a semi-abelian variety is two-step nilpotent when its ternary cosmash products vanish, or equivalently when its identity functor is quadratic. Square rings, introduced by Baues, Hartl and Pirashvili in metastable homotopy theory [1], thus extend module theory to a non-abelian setting. Similarly to abelian and semi-abelian categories, the prototypical example is the category of groups of nilpotency class at most two.

Our construction is explicit: the square ring is built from the free algebras on one and two generators, while normal forms for free objects provide concrete tools for calculations. The underlying abelian theory also remains accessible: the ordinary ring governing the abelian core is encoded in the square ring and recovered through a natural quotient construction.

The recognition theorem of Deval, Hartl and Van der Linden [2] gives a further application. Together, the two results realize every symmetric, bicocontinuous bifunctor on an abelian variety as a cosmash product of abelian objects in a category of modules over a suitable square ring. This provides a concrete algebraic framework for investigating the non-abelian structures behind tensor-like products.

Applications of our theorem and the study of the categorical properties of square rings will be presented in future work.

This is joint work in progress with Manfred Hartl and Tim Van der Linden.

Keywords: Semi-abelian category, variety of algebras, nilpotent object, cosmash product, cross-effect, polynomial functor, square ring, nearring.
Subject area: Categorical Algebra.
MSC2020: 18E13, 18C10, 08B05, 16Y30, 18E05, 20F18.

References

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[2] B. S. Deval, M. Hartl, and T. Van der Linden, Intrinsic tensor products and a Ganea-type extension of the five-term exact sequence, *J. Algebra* **710** (2027), 645–715.

[3] P. Freyd, *Abelian categories*, Harper and Row, New York, 1964, Republished in: Reprints in Theory and Applications of Categories, no. 3 (2003).

[4] L. Geenis, M. Hartl and T. Van der Linden, Two-step nilpotent varieties as categories of modules over a square ring, preprint, (2026).